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DISPLAY PANEL AND DISPLAY APPARATUS

Abstract

A display panel and a display apparatus, the display panel includes: a base plate; at least a dam provided on a side of the base plate and including a first organic layer; and an inorganic layer provided on a side of the dam away from the base plate and in contact with and connected to the first organic layer. The inorganic layer is provided with a first via extending through the inorganic layer, and an orthographic projection of the first via on the base plate is located within an orthographic projection of the dam on the base plate. The inorganic layer is provided with the first via, and the orthographic projection of the first via on the base plate is located within the orthographic projection of the dam on the base plate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Chinese Patent Application No. 202410233003.3 filed on Feb. 29, 2024, and titled “DISPLAY PANEL AND DISPLAY APPARATUS”, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of display devices, and particularly to a display panel and a display apparatus.

BACKGROUND

[0003] Organic Light-Emitting Diode (OLED) displays is also known as organic electroluminescent displays. Compared to the existing liquid crystal displays, OLED displays have a set of advantages such as self-luminescence, wide viewing angle, ultra-lightness, ultra-thinness, high brightness, low power consumption and fast response, and the response speed of the OLED displays may be 1000 times of that of the liquid crystal displays. Accordingly, OLED has become a very popular flat panel display product at home and abroad, and has a broad application prospect.

[0004] However, the process performance of the display panels needs to be improved.

SUMMARY

[0005] Embodiments of the present application provide a display panel and a display apparatus, which is aimed at improving the process performance of the display panel.

[0006] Embodiments of a first aspect of the present application provide a display panel including: a base plate; at least a dam provided on a side of the base plate and including a first organic layer; and an inorganic layer provided on a side of the dam away from the base plate and in contact with and connected to the first organic layer, wherein the inorganic layer is provided with a first via extending through the inorganic layer, and an orthographic projection of the first via on the base plate overlaps an orthographic projection of the first organic layer on the base plate.

[0007] The first aspect of the present application further provides a display panel having a display region and a non-display region provided around at least a part of the display region, and the display panel includes: a base plate; a dam provided on a side of the base plate and provided around the display region in the non-display region, wherein the dam includes a first organic layer; and an inorganic layer provided on a side of the dam away from the base plate, wherein the inorganic layer is provided with a first via extending through the inorganic layer, and the first via exposes at least a part of a region of the first organic layer.

[0008] The first aspect of the present application further provides a display panel having a display region and a non-display region provided around at least a part of the display region, and the display panel includes: a base plate; a dam provided on the base plate and provided around the display region in the non-display region, wherein the dam includes a first organic layer; and a pixel definition layer provided on the base plate and including a pixel definition portion located in the display region and a contact portion located in the non-display region, wherein the pixel definition portion is provided with a plurality of pixel openings; wherein at least a part of the contact portion covers the dam, the contact portion is provided with a first via, and an orthographic projection of the first via on the base plate overlaps an orthographic projection of the first organic layer on the base plate.

[0009] Embodiments of a second aspect of the present application provide a display apparatus

including the display panel according to any of the embodiments of the first aspect.

[0010] In the display panel according to the embodiments of the present application, the display panel includes the base plate, the dam, and the inorganic layer. The dam includes the first organic layer, and in the manufacturing process of the first organic layer, the first organic layer may release oxygen elements to form gas. The inorganic layer has a desired hermetic property, and gas flowing between the inorganic layer and the first organic layer may cause film layer peeling-off. The inorganic layer is provided with the first via, and the orthographic projection of the first via on the base plate overlaps the orthographic projection of the first organic layer on the base plate, that is, gas generated in the first organic layer may flow out of the first via, so that a phenomenon of film layer peeling-off caused by the gas remaining between the first organic layer and the inorganic layer is reduced, thereby improving the process performance of the display panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Other features, objects and advantages of the present application will become more apparent from reading the following detailed description of the non-limiting embodiments with reference to the accompanying drawings, in which the same or similar reference numerals represent the same or similar features.

[0012] FIG. 1 is a schematic structural view of a display panel according to an embodiment of the present application;

[0013] FIG. 2 is a partial enlarged schematic structural diagram at P in FIG. 1;

[0014] FIG. 3 is a cross-sectional view at A-A in FIG. 2;

[0015] FIG. 4 is a partial enlarged schematic structural view of FIG. 3;

[0016] FIG. 5 is a cross-sectional view at B-B in FIG. 2;

[0017] FIG. 6 is a cross-sectional view at C-C in FIG. 2;

[0018] FIG. 7 is a schematic structural view of a display panel according to another embodiment of the present application.

[0019] Reference Numerals: [0020] **100**: Base plate; **110**: Dam; **111**: First organic layer; **120**: First conductive line; **121**: Second via; **130**: Cushion layer; **140**: Planarization layer; [0021] **200**: Inorganic layer; **210**: Pixel definition portion; **220**: Contact portion; **230**: Pixel opening; **240**: First via; **241**: First sub-via; **242**: Second sub-via; [0022] **300**: Light-emitting layer; **310**: Light-emitting unit; [0023] **400**: Encapsulation layer; **410**: First encapsulation layer; **411**: Encapsulation portion; **420**: Second encapsulation layer; **430**: Third encapsulation layer; [0024] **500**: Isolation structure; **510**: Isolation opening; **520**: First layer; **530**: Second layer; **540**: Third layer; [0025] **700**: First electrode layer; **710**: First electrode; [0026] **800**: Second electrode layer; **810**: Second electrode; [0027] AA: Display region; NA: Non-display region.

DETAILED DESCRIPTION

[0028] Features and exemplary embodiments of various aspects of the present application will be described in detail below. Numerous specific details are set forth in the following detailed description to provide a thorough understanding of the present application. However, it will be apparent to a person skilled in the art that the present application may be practiced without some of these specific details. The following description of the embodiments is merely intended to provide a better understanding of the present application by illustrating examples of the present application. In the accompanying drawings and the following description, at least some of well-known structures and techniques are not shown so as to avoid unnecessary obscurity of the present application. In addition, size of some structures may be exaggerated for clarity. Furthermore, the features, structures, or characteristics described below may be combined in one or more embodiments by any suitable manner.

[0029] In the description of the present application, it should be noted that, unless otherwise specified, the term “a plurality of” indicates two or more; the terms “upper”, “lower”, “left”, “right”, “inner”, “outer”, and the like indicate orientations or positional relationships for facilitating and simplifying description of the present application, and do not indicate or imply that the involved devices or elements must have such a particular orientation, or must be constructed and operated in such a particular orientation, and thus should not be construed to limit the present application. In addition, the terms “first”, “second”, and the like are merely used for the purpose of description and should not be interpreted to indicate or imply relative importance.

[0030] The directional words appearing in the following description indicate directions shown in the drawings and do not limit the specific structure in the embodiments of the present application. In the description of the present application, it should be further noted that, unless otherwise clearly specified and limited, the terms “installation” and “connection” should be understood in a broad sense, for example, the “connection” may refer to a fixed, a detachable or an integrated connection; and it may refer to a direct or an indirect connection. For a person skilled in the art, the specific meaning of the terms mentioned above in the present application may be understood in accordance with specific situations.

[0031] In a display panel according to the related art, the display panel includes a base plate and a dam provided on the base plate. The dam relatively is high. Other film layers may further be provided on the dam, and the applicant has found that a problem of peeling-off easily occurs between the dam and the other film layers. Further studies by the applicant have found that this is because an organic film layer is generally provided in the dam, and the dam is formed by stacking organic film layers, but an inorganic film layer is generally provided on the dam, and the inorganic film layer has a desired hermetic property. Under a condition that the organic film layer generates gas in the manufacturing process, the gas flows between the inorganic film layer and the organic film layer, thereby causing the problem of peeling-off between the inorganic film layer and the dam.

[0032] The present application is set forth to reduce the above technical problem. For a better understanding of the present application, a display panel and a display apparatus according to embodiments of the present application will be described in detail below with reference to FIG. 1 to FIG. 7.

[0033] FIG. 1 is a schematic structural view of a display panel according to an embodiment of the present application, FIG. 2 is a partial enlarged schematic structural diagram at P in FIG. 1, FIG. 3 is a cross-sectional view at A-A in FIG. 1, and FIG. 4 is a partial enlarged schematic structural view of FIG. 3.

[0034] As shown in FIG. 1 to FIG. 4, embodiments of a first aspect of the present application provide a display panel including: a base plate **100**; at least a dam **110** provided on a side of the base plate **100** and including a first organic layer **111**; and an inorganic layer **200** provided on a side of the dam **110** away from the base plate **100** and in contact with and connected to the first organic layer **111**, in which the inorganic layer **200** is provided with a first via **240** extending through the inorganic layer **200**, and an orthographic projection of the first via **240** on the base plate **100** overlaps an orthographic projection of the first organic layer **111** on the base plate **100**.

[0035] In the display panel according to the embodiments of the present application, the display panel includes the base plate **100**, the dam **110**, and the inorganic layer **200**. The dam **110** includes the first organic layer **111**, and in the manufacturing process of the first organic layer **111**, the first organic layer **111** may release oxygen elements to form gas. The inorganic layer **200** has a desired hermetic property, and gas flowing between the inorganic layer **200** and the first organic layer **111** may cause film layer peeling-off. The inorganic layer **200** is provided with the first via **240**, and the orthographic projection of the first via **240** on the base plate **100** overlaps the orthographic projection of the first organic layer **111** on the base plate **100**, that is, gas generated in the first organic layer **111** may flow out of the first via **240**, so that a phenomenon of film layer peeling-off

caused by the gas remaining between the first organic layer **111** and the inorganic layer **200** is reduced, thereby improving the process performance of the display panel.

[0036] Optionally, the orthographic projection of the first via **240** on the base plate **100** is located within the orthographic projection of the first organic layer **111** on the base plate **100**, so that more gas can flow out of the first via **240**.

[0037] The base plate **100** may be provided in various manners, and the base plate **100** may include a substrate, and a first conductive layer, a second conductive layer, and a third conductive layer which are provided on a side of the substrate and are stacked. An insulation layer is provided between adjacent conductive layers. In an example, a pixel driving circuit is provided within the base plate **100** and includes a transistor and a storage capacitor. The transistor includes a semiconductor, a gate, a source, and a drain. The storage capacitor includes a first electrode plate and a second electrode plate. As an example, the gate and the first electrode plate may be located in the first conductive layer, the second electrode plate may be located in the second conductive layer, and the source and the drain may be located in the third conductive layer.

[0038] Optionally, the base plate **100** includes a planarization layer **140**, and a material of the planarization layer **140** includes an organic material.

[0039] Optionally, the first organic layer **111** within the dam **110** and the planarization layer **140** may be provided on the same layer and made of the same material. That is, under a condition that the planarization layer **140** is manufactured, an organic material at a location where the dam **110** is located may be retained to form the dam **110**, so that a height of the dam **110** is increased, and the manufacturing process of the display panel is simplified.

[0040] Optionally, the base plate **100** further includes a plurality of conductive layers, an insulation layer is provided between the plurality of conductive layers, and under a condition that the insulation layer is manufactured, materials in a part of dams **110** may be retained to form the first organic layer **111**, that is, the insulation layer and the first organic layer **111** are provided on the same layer and made of the same material.

[0041] Optionally, a material of the first organic layer **111** includes an organic material. The first organic layer **111** may have a relatively great thickness to ensure that the dam **110** has a relatively great height.

[0042] In some optional embodiments, the orthographic projection of the first via **240** on the base plate **100** is located within the orthographic projection of the first organic layer **111** on the base plate **100**. Accordingly, the gas generated by the first organic layer **111** can flow out of the first via **240** more quickly.

[0043] The display panel may include one or more first vias **240**, and a plurality of the first vias **240** may be provided at intervals along an extension direction of the dam **110**. By providing the plurality of the first vias **240**, gas at different locations within the organic layer can be released, and a problem of film layer peeling-off that occurs easily between the organic layer and the contact portion **220** is further reduced.

[0044] Optionally, the display panel may include one dam **110**, and the one dam **110** may be in a closed ring shape around the display region AA. Optionally, an extension direction of the dam **110** may be a direction along which the dam **110** extends around the display region AA. For example, under a condition that segments of the dam **110** located on two sides of the display region AA along a first direction extend along a second direction, an extension direction of a part of the dams **110** located on two sides of the display region AA along the first direction is the second direction. Under a condition that segments of the dam **110** located on two sides of the display region AA along a second direction extend along a first direction, an extension direction of a part of the dams **110** located on two sides of the display region AA along the second direction is the first direction.

[0045] Optionally, the display may include two or more dams **110** nested with each other. Each of the dams **110** is provided with a corresponding first via **240**, that is, each of the dams **110** is provided with a corresponding first via **240** above the organic layer, so that the problem of film

layer peeling-off that occurs easily between the organic layer and the contact portion **220** is further reduced.

[0046] In some optional embodiments, as shown in FIG. 5 and FIG. 6, a first conductive line **120** is provided on a side of the first organic layer **111** facing to the base plate **100**, the first conductive line **120** is provided with a second via **121** extending through the first conductive line **120**, the first via **121** includes a first sub-via **241**, and an orthographic projection of the first sub-via **241** on the base plate **100** at least partially overlaps an orthographic projection of the second via **121** on the base plate **100**. Accordingly, the inorganic layer **200** may cover at least a part of an inner wall surface of the first conductive line **120** facing to the second via **121**.

[0047] Optionally, the orthographic projection of the first sub-via **241** on the base plate **100** is located within the orthographic projection of the second via **121** on the base plate **100**.

[0048] In these optional embodiments, the first conductive line **120** is further provided with the second via **121**, the orthographic projection of the first sub-via **241** on the base plate **100** is located within the orthographic projection of the second via **121** on the base plate **100**, that is, a size of the first sub-via **241** is less than a size of the second via **121**, and a part of the inorganic layer **200** may extend into the second via **121** to wrap the inner wall surface of the first conductive line **120** facing to the second via **121**, so that the inorganic layer **200** may provide protection for the first conductive line **120**.

[0049] Optionally, the display panel has a display region AA and a non-display region NA provided around the display region AA, and the dam **110** may be provided in the non-display region NA and provided around the display region AA. The first conductive line **120** may intersect the dam **110** along a direction from the display region AA to the non-display region NA, or the first conductive line **120** may be provided in parallel with a part of the dam **110**.

[0050] Optionally, the first conductive line **120** may be a power signal line. For example, the first conductive line **120** is configured to transmit a low-level voltage signal ELVSS, or the first conductive line **120** is configured to transmit a driving power supply voltage signal ELVDD.

[0051] Optionally, the first conductive line **120** may be provided as a single film layer, or the first conductive line **120** may include a plurality of stacked conductive layers. For example, the first conductive line **120** includes a first sub-layer and a second sub-layer that are stacked, the second sub-layer is located on a side of the first sub-layer away from the base plate **100**, a material of the first sub-layer may be aluminium, and a material of the second sub-layer may be titanium.

Optionally, the first conductive line **120** may further include a third sub-layer located on a side of the first sub-layer away from the second sub-layer, and a material of the third sub-layer may include titanium.

[0052] Optionally, a cushion layer **130** is provided on a side of the first conductive line **120** facing to the base plate **100**, and a material of the cushion layer **130** may include an organic material. In the manufacturing process, since the organic material has desired fluidity, the first organic layer **111** above the first conductive line **120** tends to be relatively thin. In the etching process of the inorganic layer **200**, etching gas contains elemental oxygen gas, and the elemental oxygen gas may etch away the relatively thin organic layer **111**, leaving the second sub-layer exposed. The etching gas may contain elemental fluorine gas, and the elemental fluorine gas may react with the titanium to cause damage to the second sub-layer, possibly resulting in an exposed first sub-layer. Under a condition that the material of the first sub-layer includes aluminum, the aluminum may react with the oxygen element to form a greater volume of aluminum oxide, thereby causing separation of the film layers and resulting in encapsulation degradation.

[0053] In the embodiments of the present application, although the first conductive line **120** is provided with the second via **121**, the inorganic layer **200** may cover the inner wall surface of the first conductive line **120** facing to the second via **121**, that is, a region where the first conductive line **120** exists is covered by the contact portion **220**, so that the probability of the first conductive line **120** being etched and deformed or reacting with the oxygen element may be reduced, thereby

further improving the encapsulation yield.

[0054] Numbers of the first sub-vias **241** and the second vias **121** are various, for example, the display panel may include one first sub-via **241** and one second via **121**. Optionally, in some other embodiments, the display panel may include a plurality of the first sub-vias **241** and a plurality of the second vias **121**, and an orthographic projection of each of the first sub-vias **241** on the base plate **100** is located within an orthographic projection of a corresponding second via **121** on the base plate **100**. Accordingly, the inorganic layer **200** provides better protection for the first conductive line **120**.

[0055] Optionally, under a condition that the display panel includes the plurality of the first sub-vias **241**, the plurality of the first sub-vias **241** may be provided at intervals along the extension direction of the dam **110**, so that gas generated by first organic layers **111** at different locations can flow out through the first sub-vias **241**.

[0056] In some optional embodiments, the first via **240** further includes a second sub-via **242**, an orthographic projection of the second sub-via **242** on the base plate **100** is located outside an orthographic projection of the first conductive line **120** on the base plate **100**, and an area of the orthographic projection of the second sub-via **242** on the base plate **100** is greater than an area of the orthographic projection of the first sub-via **241** on the base plate **100**.

[0057] In these optional embodiments, the first via **240** may further include a second sub-via **242** that does not overlap the first conductive line **120**, a size of the second sub-via **242** is greater than a size of the first sub-via **241**, that is, the size of the first sub-via **241** is relatively small, so that the inorganic layer **200** may better cover the first conductive line **120**. The relatively great size of the second sub-via **242** can increase a rate at which the gas generated in the first organic layer **111** flows out of the inorganic layer **200**, and further reduce a problem of peeling-off caused easily between the inorganic layer **200** and the first organic layer **111**.

[0058] Optionally, the display panel may only include the second sub-via **242**, and does not include the first sub-via **241**.

[0059] As described above, the display panel includes the display region AA and the non-display region NA, the dam **110** is located in the non-display region NA, the inorganic layer **200** includes a pixel definition portion **210** located in the display region AA and the contact portion **220** located in the non-display region NA, the contact portion **220** is provided with the first via **240**, and the pixel definition portion **210** is provided with a plurality of pixel openings **230**.

[0060] In these optional embodiments, the pixel definition portion **210** is provided on the same layer as the contact portion **220**, that is, a film layer provided with the first via **240** is provided on the same layer as a film layer provided with the pixel opening **230**, and under a condition that the pixel definition layer is manufactured, a part of the material may be retained to cover the dam **110**, so that the manufacturing process of the display panel can be simplified and the manufacturing efficiency of the display panel can be improved.

[0061] In these optional embodiments, the non-display region NA includes a lower frame, and the first via **240** is provided at least in the lower frame.

[0062] Optionally, the display panel further includes a light-emitting layer **300**, and the light-emitting layer **300** includes a light-emitting unit **310** located within the pixel opening **230** to achieve light-emitting and display of the display panel.

[0063] In some optional embodiments, as shown in FIG. 5 to FIG. 7, the display panel further includes an isolation structure **500** provided on the base plate **100**, and an isolation opening **510** surrounded by the isolation structure **500**, and an orthographic projection of the isolation opening **510** on the base plate **100** at least partially overlaps an orthographic projection of the pixel opening **230** on the base plate **100**.

[0064] In these optional embodiments, by providing the isolation structure **500**, the light-emitting material can be divided into a plurality of light-emitting units **310** which are independent from each other and each located in a corresponding isolation opening **510**, so that an evaporation process of

the fine mask is omitted, and the manufacturing process of the display panel is simplified. The orthographic projection of the isolation opening **510** on the base plate **100** at least partially overlaps the orthographic projection of the pixel opening **230** on the base plate **100**, so that the light-emitting unit **310** can be located within both the isolation opening **510** and the pixel opening **230**. [0065] Optionally, the orthographic projection of the pixel opening **230** on the base plate **100** is located within the orthographic projection of the isolation opening **510** on the base plate **100**, that is, a size of the isolation opening **510** is greater than a size of the pixel opening **230**, so that more light-emitting material can fall into the pixel opening **230**.

[0066] Optionally, the base plate **100** is provided with a first electrode layer **700** including a plurality of first electrodes **710** which are provided at intervals, and an orthographic projection of each of the first electrodes **710** on the base plate **100** at least partially overlaps an orthographic projection of a corresponding pixel opening **230** on the base plate **100**, so that the first electrode **710** can drive the light-emitting unit **310** within the pixel opening **230** to emit light. Optionally, the orthographic projection of the pixel opening **230** on the base plate **100** is located within the orthographic projection of the first electrode **710** on the base plate **100**, so that a contact area between the light-emitting unit **310** and the first electrode **710** is increased.

[0067] Optionally, the display panel further includes a second electrode layer **800**, the second electrode layer **800** includes second electrodes **810** respectively located on a side of the light-emitting units **310** away from the base plate **100**, and the second electrode **810** and the first electrode **710** may interact with each other to drive the light-emitting unit **310** to emit light. Optionally, a material of the isolation structure **500** includes a conductive material, and the second electrodes **810** are in contact with and connected to the isolation structure **500**, so that the second electrodes **810** may be interconnected as an entire electrode by the isolation structure **500**.

[0068] Optionally, the display panel further includes an encapsulation layer **400** including a first encapsulation layer **410**, and the first encapsulation layer **410** includes an encapsulation portion **411** located on a side of a corresponding second electrode **810** away from the base plate **100**, so that each of encapsulation portions **411** can seal a corresponding light-emitting unit **310**, and an impact of water and oxygen intrusion on the light-emitting effect of the light-emitting unit **310** is reduced.

[0069] Optionally, the encapsulation portion **411** is located in the display region AA. The encapsulation layer **400** further includes a second encapsulation layer **420** located on a side of the first encapsulation layer **410** away from the base plate **100**, and the second encapsulation layer **420** is in contact with and connected to at least a part of the contact portion **220**. A problem can be reduced that the contact area between the first encapsulation layer **410** and the second encapsulation layer **420** is so large that the second encapsulation layer **420** has excessive fluidity, thereby improving the encapsulation performance of the second encapsulation layer **420**.

[0070] Optionally, the second encapsulation layer **420** and the contact portion **220** both extend to the non-display region NA and are in contact with and connected to each other within the non-display region NA. The first encapsulation layer **410** is located within the display region AA, the second encapsulation layer **420** and the contact portion **220** extend from the first encapsulation layer **410** to be connected to each other.

[0071] Optionally, each of the encapsulation portions **411** is located on a side of a corresponding second electrode **810** away from the base plate **100**, so that each of the encapsulation portions **411** can provide protection for the corresponding second electrode **810**.

[0072] Optionally, an orthographic projection of each of the second electrodes **810** on the base plate **100** is located within an orthographic projection of a corresponding encapsulation portion **411** on the base plate **100**, that is, a size of the encapsulation portion **411** is greater than a size of the second electrode **810**, so that the encapsulation portion **411** can provide better protection for the second electrode **810**.

[0073] Optionally, an orthographic projection of each of the isolation openings **510** on the base plate **100** is located within the orthographic projection of the corresponding encapsulation portion

411 on the base plate **100**, that is, the size of the encapsulation portion **411** is relatively great, and the encapsulation portion **411** may extend to a surface of the isolation structure **500** away from the base plate **100** to provide better encapsulation performance.

[0074] Optionally, a gap is provided between adjacent encapsulation portions **411** and is located on a side of the isolation structure **500** away from the base plate **100**, that is, a plurality of encapsulation portions **411** are independent from each other.

[0075] Optionally, a material of the inorganic layer **200** may include at least one of silicon nitride, silicon oxide, and silicon oxynitride.

[0076] Optionally, the inorganic layer **200** is a single film layer, or the inorganic layer **200** includes a plurality of stacked film layers, that is, the inorganic layer **200** may be formed by stacking the plurality of film layers.

[0077] Optionally, a film-forming condition for the inorganic layer **200** is above 150° C., films formed in the film-forming condition for the pixel definition layer **200** are more hermetically sealed than films formed in a film-forming condition for the first encapsulation layer **410**, and free bonds of the oxygen element are relatively less, so that the overflow of the second encapsulation layer **420** is buffered to a certain extent.

[0078] Optionally, a material of the first encapsulation layer **410** includes an inorganic material, that is, a material of each of the encapsulation portions **411** includes an inorganic material, so that each of the encapsulation portions **411** has a desired hermetic sealing, and is able to provide a better hermetic protection for a corresponding light-emitting unit **310**.

[0079] Optionally, a material of the second encapsulation layer **420** includes an organic material, so that the second encapsulation layer **420** may have a relatively great thickness, and flatness of the whole surface of the encapsulation layer **400** can be adjusted.

[0080] In some optional embodiments, the encapsulation layer **400** further includes a third encapsulation layer **430** located on a side of the second encapsulation layer **420** away from the base plate **100**, and the third encapsulation layer **430** in the non-display region NA extends from the second encapsulation layer **420** and is in contact with and connected to the contact portion **220**.

[0081] In these optional embodiments, the encapsulation layer **400** further includes the third encapsulation layer **430**, and the third encapsulation layer **430** in the non-display region NA extends from the second encapsulation layer **420**, that is, a size of the third encapsulation layer **430** is greater than a size of the second encapsulation layer **420**, and the third encapsulation layer **430** is in contact with and connected to the contact portion **220** outside the second encapsulation layer **420**, thereby improving the overall sealing performance of the encapsulation layer **400**.

[0082] Optionally, an edge of an orthographic projection of the second encapsulation layer **420** on the base plate **100** is located on a side of an outer edge of an orthographic projection of the contact portion **220** on the base plate **100** facing to the display region AA. That is, the edge of the contact portion **220** extends from the second encapsulation layer **420** along a direction from the display region AA to the non-display region NA, and the contact portion **220** may be hermetically connected to the third encapsulation layer **430** outside the second encapsulation layer **420**. That is, one part of the contact portion **220** is in contact with and connected to the second encapsulation layer **420**, and the other part of the contact portion **220** is in contact with and connected to the third encapsulation layer **430**, so that a contact area between the contact portion **220** and the second encapsulation layer **420** and the third encapsulation layer **430** can be increased, and the sealing effect of the encapsulation layer **400** is further improved.

[0083] Optionally, the orthographic projection of the second encapsulation layer **420** on the base plate **100** is located within an orthographic projection of the third encapsulation layer **430** on the base plate **100**. That is, the size of the third encapsulation layer **430** is greater than the size of the second encapsulation layer **420**, and the third encapsulation layer **430** can provide a longer sealing length, so that the sealing effect of the encapsulation layer **400** is further improved.

[0084] Optionally, a material of the third encapsulation layer **430** includes an inorganic material, so

that the third encapsulation layer **430** has a better sealing effect, thereby improving the overall sealing performance of the encapsulation layer **400**.

[0085] Optionally, the second encapsulation layer **420** is located on a side of the dam **110** facing to the display region AA, and the contact portion **220** and the third encapsulation layer **430** extend to a side of the dam **110** away from the display region AA.

[0086] In these optional embodiments, the second encapsulation layer **420** is located on a side of the dam **110** facing to the display region AA, the second encapsulation layer **420** is limited by the dam **110** in a region where the dam **110** is enclosed and formed, and the third encapsulation layer **430** and the contact portion **220** extend to a side of the dam **110** away from the display region AA, that is, the third encapsulation layer **430** and the contact portion **220** extend to an outer side of the dam **110**, so that the encapsulation length can be increased, and the encapsulation yield is improved.

[0087] Optionally, in a region where the dam **110** is located, the contact portion **220** is in contact with the third encapsulation layer **430**, that is, the contact portion **220** and the third encapsulation layer **430** are connected on a side of the dam **110** away from the base plate **100**, and the dam **110** is generally relatively high, so that a segment gap can be increased because of the dam **110**. The contact portion **220** is in contact with and connected to the third encapsulation layer **430** at a location with a relatively great segment gap, so that the encapsulation yield can be further improved.

[0088] The isolation structure **500** may be provided in various manners, for example, the isolation structure **500** is a single-layer structure and includes a first surface facing to the base plate **100** and a second surface away from the base plate **100**, and an orthographic projection of the first surface on the base plate **100** is located within an orthographic projection of the second surface on the base plate **100**. That is, a size of the first surface is less than a size of the second surface, so that a concave structure may be formed under the second surface, and under a condition that the light-emitting unit **310** is manufactured subsequently, the light-emitting material is partitioned into a plurality of independent light-emitting units **310** on an edge of the second surface.

[0089] Optionally, in other embodiments, the isolation structure **500** includes a first layer **520** and a second layer **530** which are stacked, the second layer **530** is located on a side of the first layer **520** away from the base plate **100**, and an orthographic projection of the first layer **520** on the base plate **100** is located within an orthographic projection of the second layer **530** on the base plate **100**. That is, a size of the first layer **520** is less than a size of the second layer **530**, so that a concave structure may be formed under the second layer **530**, and under a condition that the light-emitting unit **310** is manufactured subsequently, the light-emitting material is partitioned into a plurality of independent light-emitting units **310** on an edge of the second layer **530**.

[0090] Optionally, a material of the first layer **520** includes a conductive material and is in contact with and connected to the second electrodes **810**, so that a plurality of second electrodes **810** may be interconnected as an entire electrode by the first layer **520**.

[0091] Optionally, a material of the second layer **530** includes a conductive material, and the material of the second layer **530** is different from the material of the first layer **520**. In these optional embodiments, the material of the second layer **530** includes the conductive material, so that a distribution area of the conductive structure can be increased, and the overall resistance of the second electrodes **810** can be decreased. The material of the second layer **530** is different from the material of the first layer **520**, so that the first layer **520** and the second layer **530** with different sizes are formed using different etching rates under a condition that different materials react with the same etching solution.

[0092] Optionally, the isolation structure **500** further includes a third layer **540** located on a side of the first layer **520** facing to the base plate **100**, the orthographic projection of the first layer **520** on the base plate **100** is located within an orthographic projection of the third layer **540** on the base plate **100**. Under a condition that the isolation structure **500** is formed by performing a side etching

on the first layer **520**, the third layer **540** may have a protection effect to reduce the impact of etching on a film layer on a side of the third layer **540** facing to the base plate **100**.

[0093] In any of the above embodiments, a relative positional relationship between the isolation structure **500** and the inorganic layer **200** may be various, for example, the isolation structure **500** may be located on a side of the inorganic layer **200** away from the base plate **100**, or the isolation structure **500** may be located on a side of the pixel definition portion **210** away from the base plate **100**. Optionally, the pixel definition portion **210** is provided with an accommodation opening, and the isolation structure **500** may be located in the accommodation opening, and may be in direct contact with and connected to the base plate **100**.

[0094] The embodiments of the first aspect of the present application further provide a display panel having a display region AA and a non-display region NA provided around at least a part of the display region AA, and the display panel includes: a base plate **100**; a dam **110** provided on a side of the base plate **100** and provided around the display region AA in the non-display region NA, wherein the dam **110** includes a first organic layer **111**; and an inorganic layer **200** provided on a side of the dam **110** away from the base plate **100**, wherein the inorganic layer **200** is provided with a first via **240** extending through the inorganic layer **200**, and the first via **240** exposes at least a part of a region of the first organic layer **111**.

[0095] In the embodiments of the present application, the first via **240** exposes at least a part of a region of the first organic layer **111**, that is, at least a part of the first organic layer **111** can be exposed by the first via **240**, and the gas generated in the first organic layer **111** may leak out through the first via **240**, thereby reducing a problem of peeling-off easily caused by the presence of gas between the first organic layer **111** and the inorganic layer **200**.

[0096] The embodiments of the first aspect of the present application further provide a display panel having a display region AA and a non-display region NA provided around at least a part of the display region AA, and the display panel includes: a base plate **100**; a dam **110** provided on the base plate **100** and provided around the display region AA in the non-display region NA, wherein the dam **110** includes a first organic layer **111**; and a pixel definition layer provided on the base plate **100** and including a pixel definition portion **210** located in the display region AA and a contact portion **220** located in the non-display region NA, wherein the pixel definition portion **210** is provided with a plurality of pixel openings **230**; wherein at least a part of the contact portion **220** covers the dam **110**, the contact portion **220** is provided with a first via **240**, and an orthographic projection of the first via **240** on the base plate **100** overlaps an orthographic projection of the first organic layer **111** on the base plate **100**.

[0097] In these optional embodiments, the display panel includes the base plate **100**, the dam **110**, and the pixel definition layer. The dam **110** is provided around the display region AA, and is configured to define a position of an encapsulation material of the organic encapsulation layer **400**. The pixel definition layer includes the pixel definition portion **210** and the contact portion **220**, and the pixel definition portion **210** is provided with the pixel opening **230** to accommodate the light-emitting unit **310**, so that the display panel emits light and displays. The contact portion **220** covers the dam **110**, and the contact portion **220** is provided with the first via **240**, so that the gas generated in the first organic layer **111** flows out, thereby reducing a problem of peeling-off that occurs easily between the contact portion **220** and the first organic layer **111**.

[0098] Optionally, a material of the pixel-definition layer includes an inorganic material, and the pixel-definition layer may be provided in the same manner as the above inorganic layer **200**.

[0099] Optionally, the base plate **100** and the dam **110** are provided as described above, which is not repeated herein.

[0100] Relevant technical solutions such as the isolation structures are recited in patents 202311117143.6, 202310759370.2, 202310771124.9, 202311499823.9, 202310731471.9, 202410008807.3, 202311091555.7 and 202310707209.0, and contents thereof are incorporated into the present application by reference, and are not repeated in the present embodiment.

[0101] Embodiments of a second aspect of the present application provide a display apparatus including the display panel according to any of the embodiments of the first aspect. Since the display apparatus according to the embodiments of the second aspect of the present application includes the display panel according to any of the embodiments of the first aspect, the display apparatus according to the embodiments of the second aspect of the present application has the beneficial effects of the display panel according to any of the embodiments of the first aspect, which is not repeated herein.

[0102] The display apparatus according to the embodiments of the present application includes, but is not limited to a mobile phone, a personal digital assistant (PDA), a tablet computer, e-book, a television, an access control, a smart fixed phone, a console and other devices with a display function.

[0103] Although the present application has been described with reference to the preferred embodiments, various modifications can be made thereto and components thereof can be replaced with their equivalents without departing from the scope of the present application. In particular, various technical features described in various embodiments can be combined in any manner as long as there is no structural conflict. The present application is not limited to the specific embodiments described herein, and includes all technical solutions that fall within the scope of the claims.

Claims

1. A display panel, comprising: a base plate; at least a dam, provided on a side of the base plate and comprising a first organic layer; and an inorganic layer, provided on a side of the dam away from the base plate and in contact with and connected to the first organic layer, wherein the inorganic layer is provided with a first via extending through the inorganic layer, and an orthographic projection of the first via on the base plate overlaps an orthographic projection of the first organic layer on the base plate.
2. The display panel according to claim 1, wherein the orthographic projection of the first via on the base plate is located within the orthographic projection of the first organic layer on the base plate.
3. The display panel according to claim 1, further comprising: a first conductive line, provided on a side of the first organic layer facing to the base plate, wherein the first conductive line is provided with a second via extending through the first conductive line, the first via comprises a first sub-via, and an orthographic projection of the first sub-via on the base plate at least partially overlaps an orthographic projection of the second via on the base plate.
4. The display panel according to claim 3, wherein the orthographic projection of the second via on the base plate is located within an orthographic projection of the dam on the base plate.
5. The display panel according to claim 4, wherein the orthographic projection of the first sub-via on the base plate is located within the orthographic projection of the second via on the base plate.
6. The display panel according to claim 3, wherein orthographic projections of a plurality of the first sub-vias on the base plate are located within an orthographic projection of a same dam on the base plate; and the plurality of the first sub-vias are provided at intervals along an extension direction of the dam.
7. The display panel according to claim 3, wherein the display panel comprises a plurality of the first sub-vias and a plurality of the second vias, the plurality of the first sub-vias and the plurality of the second vias are in a one-to-one correspondence, and an orthographic projection of each of the first sub-vias is located within an orthographic projection of a corresponding second via of the plurality of the second vias.
8. The display panel according to claim 7, wherein the first conductive line comprises a power signal line; and the inorganic layer covers an inner wall surface of the first conductive line facing to

the second via.

9. The display panel according to claim 3, wherein the first via further comprises a second sub-via, and an orthographic projection of the second sub-via on the base plate is located outside an orthographic projection of the first conductive line on the base plate.

10. The display panel according to claim 9, wherein a length of the second sub-via along an extension direction of the dam is greater than a length of the first sub-via along the extension direction of the dam.

11. The display panel according to claim 9, wherein an area of the orthographic projection of the second sub-via on the base plate is greater than an area of the orthographic projection of the first sub-via on the base plate; and the second sub-via extends and is formed along the extension direction of the dam.

12. The display panel according to claim 1, wherein the display panel comprises a display region and a non-display region provided around the display region, the dam is located in the non-display region, the inorganic layer comprises a pixel definition layer comprising a pixel definition portion located in the display region and a contact portion located in the non-display region, the first via is provided in the contact portion, and the pixel definition portion is provided with a plurality of pixel openings; and a light-emitting unit is provided in the pixel opening.

13. The display panel according to claim 12, further comprising: an isolation structure provided on the base plate, an isolation opening surrounded by the isolation structure, wherein an orthographic projection of the isolation opening on the base plate at least partially overlaps an orthographic projection of the pixel opening on the base plate.

14. The display panel according to claim 13, wherein the isolation structure is located on a side of the pixel definition layer away from the base plate, or the pixel definition portion is provided with an accommodation opening, and the isolation structure is provided in the accommodation opening.

15. The display panel according to claim 13, wherein the isolation structure comprises a first layer and a second layer located on a side of the first layer away from the base plate, and an orthographic projection of the first layer on the base plate is located within an orthographic projection of the second layer on the base plate; and a material of the first layer comprises a conductive material.

16. The display panel according to claim 15, wherein the isolation structure further comprises a third layer located on a side of the first layer facing to the base plate, and the orthographic projection of the first layer on the base plate is located within an orthographic projection of the third layer on the base plate, or the orthographic projection of the first layer on the base plate overlaps the orthographic projection of the third layer on the base plate.

17. The display panel according to claim 1, wherein the display panel comprises a display region and a non-display region provided around the display region, the at least a dam comprises a plurality of the dams provided in sequence along a direction from the display region to the non-display region, and each of the dams comprises the first organic layer.

18. A display panel, wherein the display panel has a display region and a non-display region provided around at least a part of the display region, and the display panel comprises: a base plate; a dam provided on a side of the base plate and provided around the display region in the non-display region, wherein the dam comprises a first organic layer; and an inorganic layer provided on a side of the dam away from the base plate, wherein the inorganic layer is provided with a first via extending through the inorganic layer, and the first via exposes at least a part of a region of the first organic layer.

19. A display panel, wherein the display panel has a display region and a non-display region provided around at least a part of the display region, and the display panel comprises: a base plate; a dam provided on a side of the base plate and provided around the display region in the non-display region, wherein the dam comprises a first organic layer; and a pixel definition layer provided on a side of the base plate and comprising a pixel definition portion located in the display region and a contact portion located in the non-display region, wherein the pixel definition portion

is provided with a plurality of pixel openings; wherein at least a part of the contact portion covers the dam, the contact portion is provided with a first via, and an orthographic projection of the first via on the base plate overlaps an orthographic projection of the first organic layer on the base plate.

20. A display apparatus, comprising the display panel according to claim 1.
